

AI GitHub Repository Analyzer

Mini Project Report

Abstract

This project automates the analysis of public GitHub repositories using n8n workflows and the GitHub REST API. Users submit a repository URL through a webhook, and the workflow extracts repository details such as name, description, stars, forks, open issues, and default branch.

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Introduction

n8n is a workflow automation platform that connects services without extensive coding. This project demonstrates API integration, webhook handling, JavaScript processing, and response formatting.

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Problem Statement

Developers often need quick repository information. Manually browsing GitHub is repetitive. An automated workflow provides repository insights instantly.

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Objectives

- Automate repository analysis.
 - Accept GitHub repository URL.
 - Fetch data using GitHub REST API.
 - Return formatted JSON response.
 - Demonstrate workflow automation.
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 - Demonstrate workflow automation.

Technology Stack

n8n Cloud
GitHub REST API
JavaScript
Webhook
HTTP Request Node
Edit Fields Node
Respond to Webhook Node

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System Architecture

Workflow: Webhook → Code (JavaScript) → HTTP Request → Edit Fields → Respond to Webhook.

Webhook receives URL, JavaScript extracts owner/repository, HTTP Request calls GitHub API, Edit Fields formats output, Respond node returns JSON.

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Implementation

1. Configure Webhook.
2. Add JavaScript node to parse repository URL.
3. Use HTTP Request GET `https://api.github.com/repos/{owner}/{repo}`.
4. Map required fields.
5. Return response through Respond to Webhook.

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Testing

Input JSON:

```
{"url":"https://github.com/n8n-io/n8n"}
```

Expected output includes Repository, Description, Stars, Forks, Open Issues, Default Branch. During testing, the production webhook returned HTTP 200 OK.

Input JSON:

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Advantages

Easy to use, reusable workflow, real-time GitHub data, no coding for end users, scalable and extendable.

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Limitations

Works with public repositories only, subject to GitHub API rate limits, internet connection required.

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Future Scope

Add AI-generated repository summaries, contributor analytics, commit statistics, PDF export, authentication for private repositories.

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Conclusion

The project successfully demonstrates workflow automation using n8n and GitHub APIs. It reduces manual effort and provides instant repository insights through a simple webhook interface.

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References

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<https://docs.github.com/en/rest>

<https://api.github.com>